

# Artificial Intelligence Basics Course Summary

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## 1 Key Terms

### 1.1 Supervised Learning Algorithms

- **Perceptron**: A fundamental unit of a neural network, consisting of weighted inputs, a summation function, and an activation function to produce an output.

### 1.2 Search Algorithms

- **Minimax Search**: A decision rule used in game theory and decision-making, aiming to minimize the maximum possible loss for a worst-case scenario.
- **Alpha-Beta Pruning**: A search algorithm that optimizes the minimax search by eliminating branches that cannot possibly influence the final decision.

### 1.3 Artificial Intelligence Foundations

- **Turing Test**: An operational test for intelligent behavior, where a human evaluator engages in natural language conversations with both a human and a machine, attempting to distinguish them. The machine passes if the evaluator cannot reliably identify it as such.
- **Chinese Room Argument**: A thought experiment that challenges the claim that passing the Turing Test implies true understanding or consciousness.
- **Inductive Reasoning**: The process of deriving general principles from specific observations.
- **Linear Regression**: A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.
- **Model Evaluation**: The process of assessing the performance of a machine learning model.
- **Overfitting**: A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Empirical Risk Minimizer**: A machine learning approach that aims to find the model that minimizes the average loss on the training data.
- **Decision Boundary**: A line or surface that separates different classes of data points in a classification problem.
- **0-1 Loss**: A loss function in classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss**: A loss function used in the perceptron algorithm, defined as  $\max\{0, -z\}$ , where  $z = y(w^T x)$ .
- **Hinge Loss**: A loss function used in support vector machines (SVMs), defined as  $\max\{0, 1 - z\}$ , where  $z = y(w^T x)$ .

- **Logistic Loss:** A loss function used in logistic regression, defined as  $\log(1 + e^{-z})$ , where  $z = y(w^T x)$ .
- **Stochastic Gradient Descent:** An optimization algorithm that updates model parameters using the gradient calculated from a randomly selected subset of the training data.
- **Logistic Regression:** A machine learning algorithm used for binary classification that models the probability of a data point belonging to a particular class using a logistic function.

## 1.4 Linear Algebra for Machine Learning

- **Matrix:** A rectangular array of numbers, symbols, or expressions, arranged in rows and columns.
- **Linear Transform:** A function that maps vectors from one vector space to another, satisfying  $T(u+v) = T(u) + T(v)$  and  $T(cu) = cT(u)$  for all vectors  $u, v$  and scalars  $c$ .
- **Matrix Multiplication:** An operation that combines two matrices to produce a third matrix, where the element at row  $i$  and column  $j$  of the resulting matrix is the dot product of row  $i$  of the first matrix and column  $j$  of the second matrix. The inner dimensions must match.
- **Matrix Inverse:** For a square matrix  $A$ , its inverse  $A^{-1}$  is a matrix such that  $AA^{-1} = A^{-1}A = I$ , where  $I$  is the identity matrix.
- **Matrix Transposition:** An operation that swaps the rows and columns of a matrix. The transpose of matrix  $A$  is denoted as  $A^T$ .
- **Derivative:** A measure of how a function changes in response to a small change in its input. It is the slope of the tangent line to the function at a given point.
- **Partial Derivative:** The derivative of a function with respect to one of its variables, holding all other variables constant.
- **Sigmoid Function:** A mathematical function that maps any input value to an output value between 0 and 1. Often used as an activation function in neural networks.
- **Domain Decomposition:** A parallel computing technique where a problem is divided into smaller subproblems that can be solved independently on different processors.

## 1.5 Reinforcement Learning

- **Reinforcement Learning:** A machine learning paradigm where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties for its actions.

# 2 Problem Types

## 2.1 Foundational AI Concepts

- **Understanding the Nature of Intelligence:** This problem explores the definition and characteristics of intelligence, both human and artificial. It examines various perspectives on what constitutes intelligence and how it can be measured or simulated.
- **Defining Artificial Intelligence:** This problem involves formulating a precise definition of artificial intelligence, considering different perspectives and historical contexts. The challenge lies in capturing the essence of AI while accounting for its evolving nature and diverse applications.
- **Passing the Turing Test:** This problem focuses on the implications of passing the Turing Test, a benchmark for machine intelligence. It explores whether passing the test truly indicates machine intelligence or merely the ability to mimic human behavior.

- **Mitigating AI Safety Risks:** This problem examines the potential risks associated with AI systems, such as adversarial attacks and biases. It explores methods for improving the safety and reliability of AI systems.
- **Addressing Ethical Concerns in AI:** This problem explores the ethical implications of AI systems, particularly concerning bias and fairness. It examines the societal impact of AI and the need for responsible development and deployment.
- **Bridging the Gap Between Different AI Domains:** This problem explores the similarities and differences between seemingly disparate AI domains, such as game playing and speech synthesis. It aims to identify common principles and approaches that can be applied across different AI tasks.
- **Understanding the AI Stack:** This problem involves understanding the different layers and components of the AI stack, including machine learning, search, planning, computer vision, and human-AI interaction. It explores the interconnectedness of these components and their roles in building intelligent systems.

## 2.2 AI Algorithms & Optimization

- **Overcoming the Limitations of Search:** This problem addresses the computational challenges associated with exhaustive search in complex games like chess and Go. It explores the need for alternative approaches, such as machine learning, to overcome the limitations of brute-force search.
- **Developing Effective Algorithms for Complex Tasks:** This problem focuses on designing algorithms that can effectively solve complex tasks, such as fraud detection and autonomous driving. It explores the key components of such algorithms, including knowledge representation, decision-making, and generalization.

## 2.3 Linear Algebra for AI

- **Representing Linear Transformations with Matrices:** Matrices are used to represent linear transformations, which map vectors from one space to another while preserving linear combinations. This is fundamental to many AI algorithms.
- **Matrix Arithmetic:** The slide covers basic matrix operations such as addition, subtraction, scalar multiplication, and multiplication, which are essential for many AI computations.
- **Solving Systems of Linear Equations using Matrices:** Systems of linear equations can be represented and solved efficiently using matrix operations, particularly with invertible matrices.
- **Matrix Transposition:** Matrix transposition is a fundamental operation that swaps rows and columns, useful for data manipulation in AI applications.
- **Parallelizing Matrix Multiplication:** Domain decomposition is presented as a method for parallelizing matrix multiplication, improving computational efficiency in AI.

## 2.4 Calculus for AI

- **Calculus in AI:** The slide highlights the importance of calculus, specifically derivatives, partial derivatives, and finding maxima/minima/gradients, in AI algorithms.
- **Calculating the Derivative of the Sigmoid Function:** The slide shows the step-by-step derivation of the derivative of the sigmoid function using calculus rules.

## 2.5 Machine Learning Fundamentals

- **Understanding Perceptrons:** The perceptron, a fundamental building block of neural networks, is introduced, illustrating its structure and function.
- **Linear Regression:** Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression:** Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting:** Fitting a curve to a set of data points, potentially using polynomial functions of higher degrees.
- **Binary Classification:** Classifying data points into two categories (+1 or -1) using a linear model and a suitable loss function.
- **Overfitting:** The phenomenon where a model fits the training data too closely, resulting in poor generalization to unseen data.
- **Choosing a Loss Function:** Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) for binary classification, considering differentiability and computational efficiency.
- **Minimizing Loss using Gradient Descent:** Employing gradient descent or stochastic gradient descent to iteratively update model parameters and minimize the chosen loss function.

## 2.6 Game AI

- **Understanding the Differences Between Game Types:** This problem involves comparing and contrasting different game types, such as chess and soccer, to understand the unique challenges they pose for AI algorithms. It explores how the complexity and dynamics of a game influence the design of AI systems.
- **Understanding the Mechanics of Game Playing Systems:** This problem explores the fundamental mechanisms behind AI systems designed to play games, including knowledge representation, action selection, and strategy development.
- **Analyzing the Success of AlphaGo:** This problem involves analyzing the AlphaGo system, highlighting its use of deep learning and search techniques, and understanding its success in mastering the game of Go.

## 2.7 Computer Vision Challenges

- **Addressing the Challenges of Computer Vision:** This problem investigates the difficulties in developing computer vision systems that can reliably interpret and understand images. It highlights the gap between initial expectations and the ongoing challenges in this field.

# 3 Algorithm Solutions

## 3.1 Linear Algebra

- **Matrix Addition:**  $A + B = C$ , where  $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication:**  $k \cdot A = B$ , where  $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication:**  $C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$
- **Matrix Inverse:**  $AA^{-1} = A^{-1}A = I$
- **Solving Linear Equations using Matrix Inverses:**  $Ax = b \Rightarrow x = A^{-1}b$

## 3.2 Machine Learning Fundamentals

## 3.3 Artificial Neural Networks

- **Derivative of Sigmoid Function:**  $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1+e^{-x})^2}$
- **Perceptron Algorithm:** Iteratively update the weight vector  $w$  based on misclassified data points. If  $\text{sign}(w^T x_n) \neq y_n$ , then  $w \leftarrow w + y_n x_n$ .

## 3.4 Game Playing AI

- **Minimax Search:** A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha-Beta Pruning:** An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.